

skilled person to reliably achieve, without undue burden, the technical effect of expression in any plant cell of any plant structural gene under the control of any plant promoter. Further cases where more than one example was required can be found in this chapter [II.C.7.4](#).

In [T.19/90](#) (OJ 1990, 476) the claimed invention was defined by the incorporation of an activated oncogene sequence into the genome of non-human mammalian animals in general. The examining division refused the application on the grounds that in the light of the differences among different animals, it could not be assumed that the sole example given – mice – could be extended to all other non-human mammalian animals and the claims were thus unrealistically broad. The board disagreed.

In [T.636/97](#) it was emphasised that it is a fundamental principle of patent law that a claim can validly cover broad subject-matter, even though the description of the relevant patent does not enable every method of arriving at that subject matter to be carried out. Otherwise no dominant patent could exist, and each developer of a new method of arriving at that subject matter would be free of earlier patents. In [T.694/92](#) (OJ 1997, 408) the board held that, where an invention relates to the actual realisation of a technical effect anticipated at a theoretical level in the prior art, a proper balance must be found between, on the one hand, the actual technical contribution to the state of the art by said invention, and, on the other hand, the terms in which it is claimed, so that, if patent protection is granted, its scope is fair and adequate. The board highlighted the interrelation between the requirements of [Art. 84](#), [83](#) and [56 EPC 1973](#). See also [T.187/93](#).

In [T.2249/16](#) the contested decision had found fault with the claims on the grounds that they covered an extremely large number of possible compounds or methods and were also broad in comparison with the examples. The board held that this was not enough alone to establish insufficient disclosure.

On broad claims, see in particular chapter [II.C.7.3](#) below, which deals with antibodies defined by functional features.

7.2. Level of disclosure required for medical use – plausibility

7.2.1 Principles established by the case law

The case law reported below includes the landmark decisions establishing the applicable principles and examples of decisions subsequently taken in specific cases that recapitulate and apply those principles.

The boards of appeal have recognised that in the context of the requirement of sufficiency functional features require particular attention, as such features are defined by means of an effect that has to be achievable (see [G.1/03](#), point 2.5.2 of the Reasons, cited by [T.2015/20](#)).

According to the established case law of the boards of appeal, attaining the claimed therapeutic effect is regarded as a functional technical feature of claims relating to a further

medical use. In order to meet the requirement of sufficiency of disclosure of Art. 83 EPC, the therapeutic efficacy of the composition and dosage regime for the claimed therapeutic indication must therefore be credible (principle reiterated in e.g. T 421/14 (dosage regime involving twice-daily treatment – multiple sclerosis), where the claims related to a further medical use).

The decision in T 1959/15, in which granted claim 1 was a second medical use claim in the format of a purpose-restricted product claim pursuant to Art. 54(5) EPC, also helpfully summarises the applicable notions and reasoning (point 4.2 of the Reasons): According to Art. 54(5) EPC, patentability is not excluded for substances or compositions comprised in the state of the art for a specific use in a method referred to in Art. 53(c) EPC, provided that such use is not comprised in the state of the art. When a technical effect (which, in the case of a second medical use claim, is the therapeutic effect) is a feature of a claim, whether this effect is achieved by substantially all embodiments covered by the claim is a question of sufficiency of disclosure. Hence, because the subject-matter of second medical use claims is commonly limited to a known therapeutic agent for use in a new therapeutic application, it is usually only necessary that the patent renders it plausible that the known therapeutic agent (i.e. the product) is suitable for the claimed therapeutic application (i.e. the purpose: the technical effect).

T 1868/16 also stated that it is established jurisprudence of the boards of appeal that, for Art. 83 EPC to be complied with, unless this is already known to the skilled person at the priority date, the application must disclose the suitability for the claimed therapeutic use (see T 609/02, point 9 of the Reasons; T 433/05, point 28 of the Reasons; T 801/06, point 25 of the Reasons). Clinical data are not always required. Mere verbal statements are however not enough. The patent application must provide some information in the form of, for example, experimental tests to the avail that the claimed compound has a direct effect on a metabolic mechanism specifically involved in the disease. In vitro examples may be sufficient, if for the skilled person they directly and unambiguously reflect the therapeutic application. Post-published evidence may be taken into account, but only to back-up the findings in the application (cf. T 609/02, point 9 of the Reasons).

In T 609/02 the board pointed out that where a therapeutic application is claimed in the form allowed by the Enlarged Board of Appeal in G 5/83 (OJ 1985, 64), i.e. in the form of the use of a substance or composition for the manufacture of a medicament for a defined therapeutic application, attaining the claimed therapeutic effect is a functional technical feature of the claim (see G 2/88, OJ 1990, 93, and G 6/88, OJ 1990 114, for non-medical applications). As a consequence, under Art. 83 EPC, unless this is already known to the skilled person at the priority date, the application must disclose the suitability of the product to be manufactured for the claimed therapeutic application (summaries of the points established by these decisions as regards claimed therapeutic effects can be found in, for example, T 2571/12, point 5.2 of the Reasons; T 1437/07, reported below, point 37 of the Reasons; T 421/14, point 2.3 of the Reasons; T 2015/20, which contains some very detailed and instructive findings on the point; T 395/18, point 4.3. of the Reasons; and T 899/14, points 2.4 and 2.5 of the Reasons). In decisions such as T 2181/08, T 338/10, T 1685/10, T 943/13 and T 2059/13, the boards have cited the approach taken in T 433/05 and T 609/02 and applied it to the specific cases before them.

The board in T.814/12 held that it is established case law of the boards of appeal for a medical use claim to fulfil the requirements of Art. 83 EPC, unless this is already known to the skilled person at the priority date, that the patent has to disclose the suitability of the product to be manufactured for the claimed therapeutic application. A claimed therapeutic application may be proven by any kind of evidence as long as it reflects the therapeutic effect on which the therapeutic application relies (T.814/12, referring to T.609/02 in particular). In T.814/12 the board considered that by analogy the same requirements of Art. 83 EPC applied for diagnostic use claims.

In T.895/13 of 21 May 2015 the board stated that, pursuant to decision T.609/02, attaining the claimed therapeutic effect was a functional technical feature of a claim drawn up in the Swiss-type form. In its view, the same principle applied to purpose-related product claims drawn up in accordance with Art. 54(5) EPC. Accordingly, the therapeutic effect provided by the claimed subject-matter was to be examined in the context of the assessment of sufficiency of disclosure (Art. 83 EPC) (see G.1/03, point 2.5.2 of the Reasons) and not, as in the decision under appeal, in the context of the assessment of inventive step.

7.2.2 Evidence of therapeutic effect

The question is whether or not the skilled person, having regard to the disclosure of the patent and the common general knowledge at the relevant date of the application, would have considered that the compounds referred to in the claim are suitable to achieve the therapeutical effect (see T.609/02, point 9 of the Reasons). Or, in other words, whether it was plausible (or, in yet other words, whether it was credible) that the therapeutic effect could be achieved by the claimed composition (as recapitulated in T.966/18).

Either the application must provide suitable evidence for the claimed therapeutic effect or it must be derivable from the prior art or common general knowledge. The disclosure of experimental results in the application is not always required to establish sufficiency, in particular if the application discloses a **plausible** technical concept and there are no substantiated doubts that the claimed concept can be put into practice (T.950/13 citing T.578/06).

A claimed therapeutic application may be proven by any kind of evidence as long as it reflects the therapeutic effect on which the therapeutic application relies (T.814/12, referring to T.609/02 in particular).

a) Plausibility of therapeutic effect

If a therapeutic application is to be accepted as sufficiently disclosed, the application or the patent, respectively, and/or the common general knowledge has to provide some information rendering it technically **plausible** for the skilled person that the claimed compounds can be applied for the claimed therapeutic use (T.1599/06 citing T.609/02).

The requirement of sufficiency of disclosure is considered to be fulfilled with respect to a claim to a second medical use if the disclosure in the patent or the common general knowledge enables the skilled person to obtain the compound to be applied and to apply

it, and if there is evidence that the intended therapeutic effect can be achieved (T.1437/07 – Botulinum toxin). The board in that case also stated that just because a patent disclosed an effect which had not in reality been achieved, there was no reason – in the absence of convincing evidence that the effect could not be achieved – for it to doubt that the effect could be achieved (point 38.1 of the Reasons).

The board in T.899/14 decided that, contrary to the view taken by the appellant (patent proprietor), it was not sufficient merely to describe an administration regimen to be followed, without any evidence of the therapeutic efficacy of the proposed treatment. The appellant had not argued that it was common general knowledge but instead **relied on the examples**. The question was thus whether the information presented in the examples could render the alleged efficacy credible, or at least establish its initial plausibility. The board also noted that while results of clinical trials or animal studies were not always necessary to establish sufficiency of disclosure, a **mere verbal statement** in the application was not enough to establish even the **initial plausibility** of an alleged therapeutic benefit.

Having reiterated the applicable principles (point 4.2 of the Reasons), the board in T.1959/15, where granted claim 1 was a second medical use claim in the format of a purpose-restricted product claim pursuant to Art. 54(5) EPC, found that the patent in this specific case, besides amounting to an invitation to perform a research programme and thus an undue burden, showed an effect (on cardiomyopathy) for only two exemplary compounds having the desired capability. Since these two compounds were structurally very similar, it was not plausible that all possible (structurally distinct) compounds having this capability would have the same effect.

b) Consequences of plausibility for whether post-published documents considered

In T.184/16 the board recalled that a precondition for taking into account post-published evidence to demonstrate a certain effect was that it was already plausible at the filing date that said effect was obtained (see T.488/16, T.1329/04 and T.433/05). Plausibility was acknowledged, and post-published evidence was taken into account, for example in cases where there were no "prima facie serious doubts" about plausibility (T.108/09, T.1760/11 of 13 November 2012, T.919/15; by contrast, in T.1329/04 there were prima facie serious doubts). In the case in hand, the application as filed did not contain any experimental evidence as regards the disputed plausibility, i.e. the plausibility of the claimed compounds being SGLT2 inhibitors. It was thus necessary to determine whether plausibility could be acknowledged in view of the common general knowledge and the prior art. The board considered it plausible that the therapeutic effect was indeed obtained. Post-published evidence D4 (comparative examples filed by the respondent/patent proprietor) could be taken into account. The fact that SGLT1 inhibition might contribute to this effect as well and was not tested in D4 was not relevant. Furthermore, the appellant (opponent) bearing the burden of proof for its assertion, the board could not conclude, in the absence of any such evidence, that compounds with large substituents were not suitable to obtain the therapeutic effect defined in claim 12. Sufficiency of disclosure was considered to be satisfied.

In **T. 391/18** Claim 1 of the main request was directed to a combination of active ingredients (TMC278 and a NRTI (nucleoside or nucleotide reverse transcriptase inhibitor)) for treating HIV infection in a regime of once-daily administration. The patent did not contain any evidence on combinations of TMC278 with NRTIs. The **initial plausibility** – in view of the common general knowledge – could **not be equated with** meeting the requirement of Art. 83 EPC, but nevertheless allowed the board to consider post-published evidence. Post-published document D26 (report with clinical trials for authorisation) was then considered but appeared insufficient to make credible that every possible combination of TMC278 with NRTIs that were therapeutically effective by once-daily administration would be suitable for treating HIV in a once-daily dosage regime. To find suitable combinations and their corresponding doses among all the possibilities covered by claim 1, carrying out an undue amount of research was needed. In the **field of pharmaceutical combinations, drug-drug interactions** need to be assessed for each drug combination to find whether and at which dose the combination is therapeutically effective. Such an assessment involves clinical studies which cannot be considered routine tests. Even if, as argued by the respondent (patent proprietor), the number of NRTIs suitable for once-daily administration were not particularly high, the research required would go far beyond what may be seen as routine testing. Therefore, there existed serious doubts substantiated by verifiable facts. But claim 1 of auxiliary request 24, limited to encompass the combination tested in D26 met Art. 83 EPC.

In ex parte case **T. 2015/20** the examining division found that it was not plausible that acclidinium bromide was suitable for treatment of asthma. The application only presented experimental results concerning treatment of COPD (chronic obstructive pulmonary disease), whereas it was part of the common knowledge, as presented by D5, that COPD and asthma were distinct diseases with different mechanisms involved. The board observed that the information in D5 did not cast doubt on the statement in the application about the effectiveness of treatment. Concerning the post-published document D6, it merely warned that the use of "Duaklir Genuair" in asthma had not been officially authorised, which was per se not a ground for any serious doubts regarding the claimed utility of acclidinium in the treatment of asthma. In conclusion, no serious doubts could support the objection of lack of sufficient disclosure. Neither **T. 609/02** nor the later jurisprudence signalled a deviation from the established jurisprudence, in particular with respect to the precondition of serious doubts for a convincing argument of lack of sufficiency. The board took the view that in the present case the defined utility of acclidinium in treatment of asthma did not go against any prevailing opinion in the prior art. In this context the board considered the statement in the application, that the treatment of respiratory disorders, particularly asthma and COPD, with acclidinium was most effective upon administration by inhalation in a dosage of about 400 µg metered nominal dose to represent a significant technical teaching, which was far from an invitation to perform a research programme and which did not prima facie lack plausibility. This teaching was as such **falsifiable**, in the sense that it was open to challenge, and was therefore considered to represent information in the form of a specific technical contribution which went beyond some insufficient verbal statement. The board held that sufficiency was therefore not to be denied following its assessment that no serious doubts had come about with respect to the defined utility. The board turned to the examination of inventive step itself, considering in this respect also the question of plausibility.